Deep Learning for American Sign Language Gesture Classification

DS 6050

Bela Barton, Natalie Seah, Erin Siedlecki

Motivation

Who is affected?

- 250k–500k people in the U.S. rely on ASL (James et all., 2022)

The Problem

- Most hearing individuals do not understand ASL → communication gap (healthcare, education, workplace)
- Human interpreters are costly & not always available (Zulu et al., 2017)

Project Objective

- Train deep learning models to classify ASL hand gestures from images
- Predict letters (A–Z) and digits (0–9)

Related Work

Alsharif et. al (2023) – Deep Learning Technology to Recognize American Sign Language Alphabet

- Compared 5 CNN/ViT architectures on 87k static ASL images
- ResNet-50 achieved 99.98% accuracy; ViT lagged at 88.59% due to input resizing losses
- Establishes CNN transfer learning as the baseline; motivates our inclusion of ResNet-18
- Limitation: alphabet-only dataset; our work extends to alphanumeric (A-Z, 0-9) with ASL-HG

Kasapbasi et al. (2022) – DeepASLR: CNN-Based American Sign Language Recognition System

- Built ASLA dataset under variable lighting and capture distances; contrast to clean benchmark sets
- CNN achieved 99.38% test accuracy and generalized well across multiple datasets
- Dataset diversity matters as much as model architecture for real-world generalization
- Directly motivates our use of ASL-HG (raw images) with varied backgrounds, lighting, and skin tones

*Many prior works use datasets with highly similar images, which can lead to artificially high accuracy. Our project aims to address this by using more diverse data to better evaluate generalization.

Dataset

Overview

- ASL Hand Gesture Dataset (ASL-HG)
- 36,000 images: 36 classes (A–Z, 0–9)
  - Balanced across classes
- Released in 2 forms: raw images and MediaPipe preprocessed
  - Raw images were used

Data Collection

- 10 volunteers (subjects) in Dhaka, Bangladesh
- Each performs 100 images per class

Technical Approach

INPUT The inputs of this task are RGB images of hand gestures that represent an ASL letter (A-Z) or digit (0-9)

MODEL DESIGN

Model 1: Baseline CNN from Scratch

- Custom CNN (4 conv blocks, 2 conv layers each)
- BatchNorm + ReLU + Max Pooling
- Dropout (regularization)
- Global Average Pooling
- Fully connected layers (256 → 128 → number of classes)

Model 2: EfficientNet-B0

- EfficientNet-B0 transfer learning (pretrained on ImageNet)
- Compound scaling (depth, width, resolution)
- Full fine-tuning (no frozen layers)
- Final classification layer replaced

Model 3: ResNet-18

- ResNet-18 transfer learning (pretrained on ImageNet)
- Residual (skip) connections
- All layers frozen except for layer4 + final classification layer (partial fine-tuning)

OUTPUT Probability distribution over the 36 classes and the predicted label indicated by the class with the highest probability

Experimental Setup

Train/Test/Val Split

- 100 images per volunteer per class
- **Train:** volunteers 0-6 (25,200 images)
- **Validation:** volunteer 7 (3,600 images)
- **Test:** volunteers 8-9 (7,200 images)
- Data split by volunteer to avoid data leakage and allow for better generalization

Data Augmentation

- Random rotation (±15°)
- Random translation (up to 10%)
- Color jitter (brightness/contrast ±30%, saturation ±20%)
- Gaussian blur
- Random grayscale (10%)
- Random perspective transformation
- Random horizontal flip (mirror hands)

Models

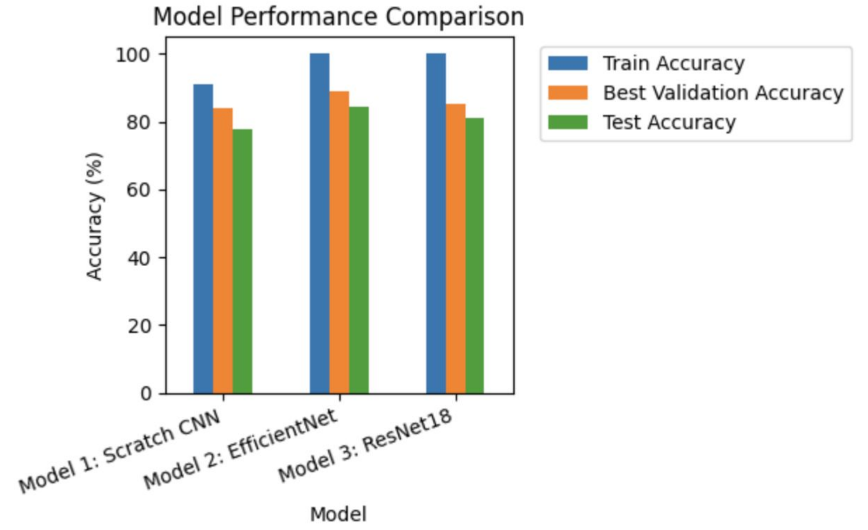
- **Model 1:** Baseline CNN (from scratch)
- **Model 2:** EfficientNet-B0 (transfer learning)
- **Model 3:** ResNet-18 (transfer learning)

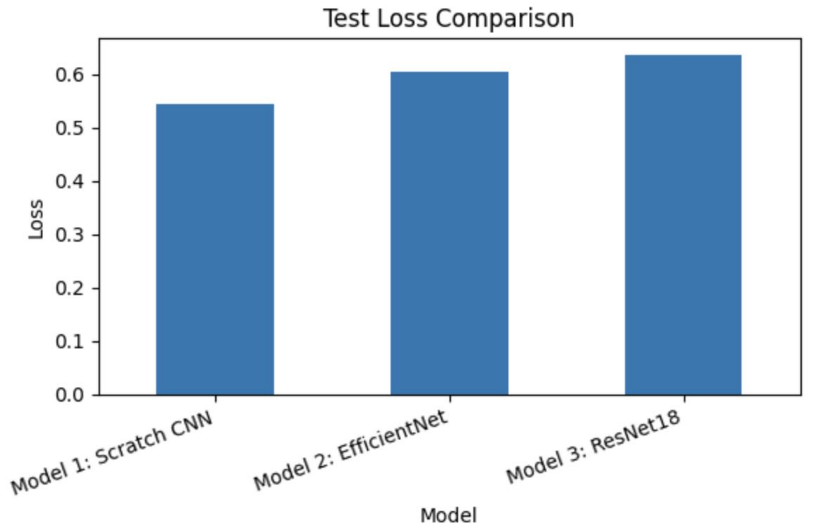
Evaluation

- **Metrics:** cross-entropy loss, training/validation/test accuracy
- Compare performance between all three models

Experimental Results

Model	Train Accuracy	Best Validation Accuracy	Test Accuracy	Test Loss
Model 1: Scratch CNN	91.05	83.89	77.93	0.5454
Model 2: EfficientNet	99.95	88.97	84.36	0.6059
Model 3: ResNet18	99.92	85.03	81.19	0.6355

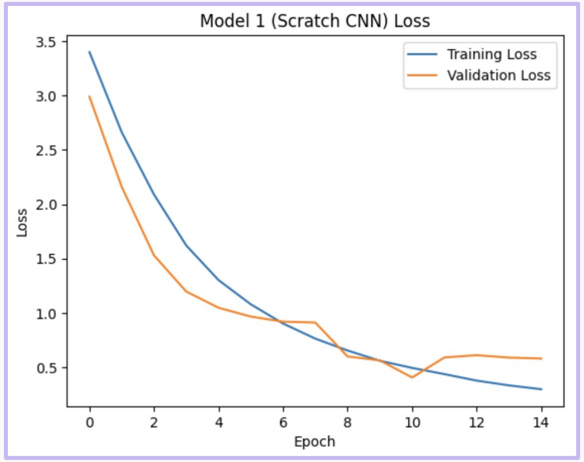




Experimental Results

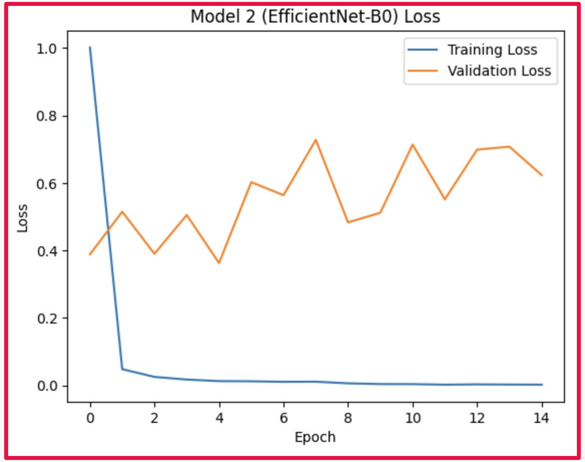
1

Baseline Model (CNN from Scratch)



2

EfficientNet-B0



3

ResNet-18

Conclusions

1 Baseline CNN (from scratch)

- Strong, simple baseline model for performance comparison
- Achieved the lowest test loss (0.5454), but also the lowest test accuracy (77.93%)
- Smaller train-test gap indicating less overfitting than other models
- Suggests better calibration, but worse overall performance

2 EfficientNet-B0

- Achieved the best test accuracy (84.36%)
- Train accuracy is nearly perfect, indicating that the model overfits
- Higher test loss than baseline model, suggesting more overconfident incorrect predictions
- Overall best performance among the three models

3 ResNet-18

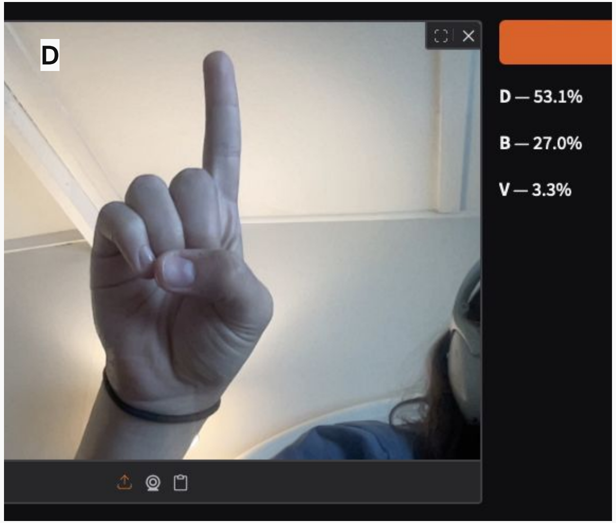
- Improved over the baseline model, but performs worse than EfficientNet
- Achieved the highest test loss (0.6355)
- Largest train-test gap, indicating strong overfitting
- Transfer learning improved performance, but overfitting persists

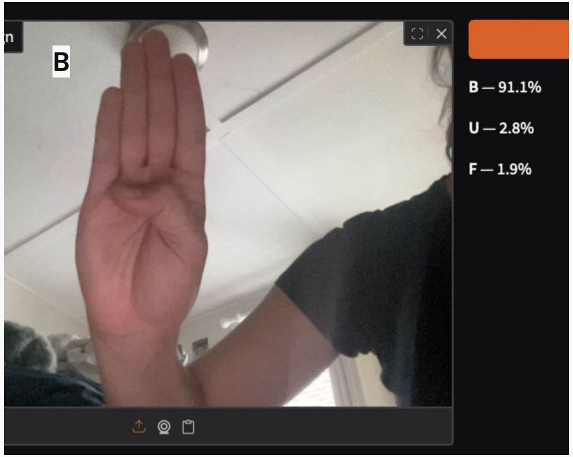
Overall: All three models exhibit overfitting; while transfer learning improves accuracy, indicating limited generalization. Our results likely reflect the limited variety of images in our dataset. These findings further illustrate the need to develop generalizable ASL gesture recognition models in an effort to reduce the communication gap. Future work could explore more diverse datasets or video-based models.

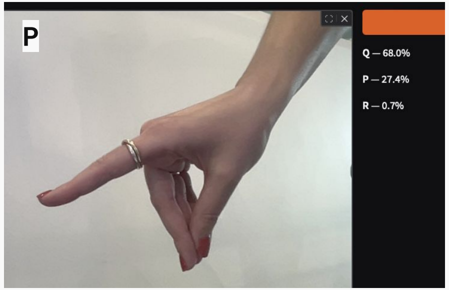
Our Application

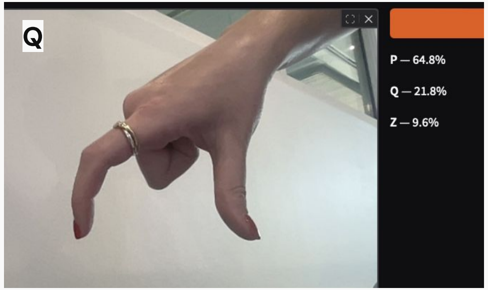
What It Does

- Text → Fingerspelling Video
  - Enter a sentence and get a fingerspelling video (A–Z, 0–9).
- **Image → Sign Classifier**
  - Upload a hand sign photo and the model returns top 3 predictions









References

Alsharif, Bader, et al. “Deep Learning Technology to Recognize American Sign Language Alphabet.” *Sensors (Basel, Switzerland)*, U.S. National Library of Medicine, 19 Sept. 2023, pmc.ncbi.nlm.nih.gov/articles/PMC10535774/.

Anthropic. (2026). Claude [Large language model]. https://claude.ai

James, T. G., McKee, M. M., Sullivan, M. K., Ashton, G., Hardy, S. J., Santiago, Y., Phillips, D. G., & Cheong, J. (2022). Community-engaged needs assessment of Deaf American Sign Language users in Florida, 2018. Public Health Reports, 137(4), 730–738. https://doi.org/10.1177/00333549211026782

Kasapbaşı, A., Ahmed Elbushra, A. E., Al-Hardanee, O., & Yılmaz, A. (2022). *DeepASLR: A CNN based human computer interface for American Sign Language recognition for hearing-impaired individuals. Computer Methods and Programs in Biomedicine Update, 2*, 100048. https://doi.org/10.1016/j.cmpbup.2021.100048

OpenAI. (2026). ChatGPT [Large language model]. https://chat.openai.com/

Pranto, M. F. I., Islam, M. R., Akbor, M. A., Ghosh, N., Alam, M. R., Chaki, S., & Islam, M. M. (2026). *A comprehensive image dataset of American Sign Language hand gestures. Data in Brief, 65*, 112492. https://doi.org/10.1016/j.dib.2026.112492

Zulu, T., Heap, M., & Sinanovic, E. (2017). The cost and utilisation patterns of a pilot sign language interpreter service for primary health care services in South Africa. PLOS ONE, 12(12), e0189983. https://doi.org/10.1371/journal.pone.0189983